W-F-785E
August 31, 1978
SUPERSEDING
Fed. Spec. W-F-785D
May 29, 1969

FEDERAL SPECIFICATION

# FURNACES, HEAT-TREATING, ELECTRIC, HORIZONTAL BOX TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers horizontal box type, natural atmosphere, heat-treating furnaces of the electrically heated, forced air convection, and radiation design.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The furnaces shall be of the following types and classes as specified (see 6.2.1).

Type I - Forced convection design.
  Class A - 1250 degree F operating temperature.
  Class B - 1400 degree F operating temperature.
Type II - Radiation design.
  Class A - 2000 degree F operating temperature.
  Class B - 2500 degree F operating temperature.
  Class C - 2750 degree F operating temperature.



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1.2.2 Sizes. The size of the furnace chamber shall be as specified (see 6.2.1). The minimum size of the furnace chamber shall be not less than 6 inches in width x 12 inches in length x 6 inches in height. Size changes shall increase in increments of 3 inches in width or height and 6 inches in length up to a chamber size of 18 W x 30 L x 18 H. Chambers larger than 18 W x 30 L x 18 H shall increase in increments of 6 inches in any dimension as required. The inside dimensions of the heating chamber shall be not less than the dimensions specified. The door opening size shall be equal to the width and height dimensions specified for the size furnace.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-F-3296 - Forges, Furnaces, and Ovens (exclusive of Space Heating and Cooking), Packaging of.

Military Standard:

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.



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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

| | |
|---|---|
| Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments | Occupational Safety and Health Administration Standards |

NATIONAL BUREAU OF STANDARDS (NBS)

H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

JOINT INDUSTRIAL COUNCIL (JIC)

EGP-1 - Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Joint Industrial Council, 2139 Wisconsin Avenue, NW, Washington, DC 20007.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C96.1 - Temperature Measurement Thermocouples

(Applications for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete furnace for first article inspection and approval (see 4.2 and 6.4).

3.2 Design. The furnace shall be new and of the type normally furnished by manufacturers of this type of equipment. The furnace shall be of the horizontal box design, having a hearth to support the charge and a single opening located in the end of the furnace for loading and unloading the charge in a horizontal plane. Unless otherwise specified, the furnace shall be floor mounted. When required (see 6.2.1), the furnace shall be bench mounted.



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3.2.1 Type I. Type I furnace shall employ the forced air convection method of heating. Design and construction shall be such that the charge shall be heated by forced air convection. The air shall be heated in an area separated from the charge and blown into the work chamber by one or more fans. The charge shall not be exposed to the radiant heat emitted by the heating elements.

3.2.2 Type II. The Type II furnace shall employ the radiation method of heating with the charge directly exposed to the radiant heat emitted by the electric heating elements.

3.2.3 Safety and health requirements. Each furnace shall be provided with appropriate safety features and devices for protection of operating personnel and equipment. Unless more stringent requirements are specified by the procuring activity, the furnace shall conform to safety and health standards promulgated under OSHA which are applicable to the furnace. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

3.2.4 Material. All material used in the fabrication of the furnace and related equipment shall be of the quality necessary to produce a furnace to meet the requirements described herein. Materials that are exposed to the internal environment of the furnace chambers shall be compatible with the atmosphere and temperature conditions they will encounter.

3.2.5 Reclaimed materials. The furnace may contain reclaimed materials provided such material will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the furnace. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

3.2.6 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack-seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material or equipment. Lubrication points shall be readily accessible, and identified in the operators manual with type of lubricant and instructions for application.

3.2.7 Accessibility. All parts subject to wear, breakage, or distortion, and all parts which require periodic maintenance shall be accessible for adjustment and replacement, as applicable.



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3.2.8 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or equipment.

3.3 Construction. The furnace shall be constructed so that when installed and connected to the specified power source, it can be used for any prupose described herein. All parts and accessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the furnace.

3.3.2 Welding, brazing, or soldering. Welding, brazing, and soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the furnace shall be painted with heat resisting paint in accordance with the manufacturer's standard commercial practice. Physical hazards on the furnace shall be color coded in accordance with 29 CFR 1910.144.

3.3.6 Threads. All threaded parts shall conform to Handbook H28.

3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.4 Electrical equipment. The furnace shall be furnished complete with all accessory electrical motors, contactors, controls, relays, transformers, signal lights, and switches prewired where possible for operation upon connection to source of power. All power operated, manually adjusted and automatic controls shall operate on and shall be wired for, 115 volts, 60 cycle current. All electrical equipment and wiring shall conform to JIC EGP-1.



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3.4.1 Power supply. The furnace shall be designed for operating from the power source specified (see 6.2.1).

3.4.2 Motors. Motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volt, initially wired to operate on the voltage specified for the furnace.

3.4.3 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with means to prevent loss of tightness.

3.4.4 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall be in accordance with the requirements of MIL-STD-461.

3.4.5 Wiring data plates. When dual voltage furnaces are supplied, metal data plates shall be furnished showing the wiring arrangement for 230- and 460- volt operation. To facilitate safe and rapid changeover from power sources of one voltage to another, plates showing heating element connections, circulating fan motor connections, and control circuits step-down transformer connections for 230 and 460 volts shall be conspicuously placed at or near the terminal boards or panels where these components are connected.

3.5 Performance. Each furnace shall be capable of operating continuously with a maximum charge at any temperature within its operating range.

3.5.1 Operating temperature range.

3.5.1.1 Type I, class A. The minimum operating temperature for the type I, class A furnace shall be not higher than 600°F and the maximum operating temperature shall be not less than 1250°F.

3.5.1.2 Type I, class B. The minimum operating temperature for the type I, class B furnace shall be not higher than 600°F, and the maximum operating temperature shall be not less than 1400°F.

3.5.1.3 Type II, class A. The type II, class A furnace shall operate with maximum efficiency within the temperature range of 1400 to 2000°F. The minimum operating temperature shall be not higher than 600°F, and the maximum operating temperature shall be not less than 2000°F.



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3.5.1.4 Type II, class B. The type II, class B furnace shall operate with maximum efficiency within the temperature range of 1400 to 2500°F. The minimum operating temperature shall be not higher than 600°F, and the maximum operating temperature shall be not less than 2500°F.

3.5.1.5 Type II, class C. The type II, class c furnace shall operate with maximum efficiency within the temperature range of 1400 to 2750°F. The minimum operating temperature shall be not higher than 1000°F, and the maximum operating temperature shall be not less than 2750°F.

3.5.2 Temperature uniformity.

3.5.2.1 Type I. The type I furnace, when charged and temperature stabilized, the temperature variation from chamber control point to any other point within the heating chamber shall not vary more than ± 10°F. The heating chamber shall be defined as the space equaled to the inside dimensions specified for the size furnace. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within its operating temperature range.

3.5.2.2 Type II. The type II furnace, when charged and temperature stabilized, the temperature variation from chamber control point to any other point within the effective working area shall not vary more than ± 10°F. The effective working area of the type II furnace may be defined as being one sixth less than each of the actual inside dimensions specified for the size furnace. Each type II furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within the maximum efficiency operating temperature range specified.

3.5.3 Recovery rate and heating capacity. Recovery rate and heating capacity shall be in accordance with 3.5.3.1 and 3.5.3.2 for the type and class furnace specified. Hearth area for heating capacity refers to the whole hearth.

3.5.3.1 Type I. The type I, class A and class B furnace shall have a recovery rate of not less than 60 pounds of steel for each square foot of hearth area per hour, to the maximum temperature specified for the class furnace. The maximum capacity of the type I furnace shall be not less than 130 pounds per square foot of hearth area.

3.5.3.2 Type II. The type II, class A furnace shall have a recovery rate equivalent to not less than 60 lbs of steel for each square foot of hearth area per hour to a temperature of 2000°F. The type II, class B and class C furnace shall have a recovery rate of not less than 50 lbs of steel for each square foot of hearth area per hour to a temperature of 2150°F. The maximum capacity of the type II, class A furnace shall be not less than 100 lbs/sq foot at 2000°F, class B, 80 lbs/sq foot at 2500°F, class C, 60 lbs/sq foot at 2750°F.



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3.6 Components and systems.

3.6.1 Frame. The furnace shall have a structural steel frame and shall provide a stable support for the work chamber and a maximum load. If legs are used, they shall be braced and provided with bearing plates. Floor mounted furnaces with work chambers up to 36 inches wide shall have a working height above the floor of not less than 30 inches. Furnaces with work chambers over 36 inches wide shall have a working height above the floor of not less than 18 inches. Furnaces that weigh 2000 pounds or more shall have provisions for forklift handling.

3.6.2 Shell. The outer shell of the furnace shall be constructed of sheet steel, properly reinforced and securely welded or riveted at all joints to provide a rigid structure. The walls shall be constructed in such a manner that the use of through fastening devices which conduct heat to the outer shell will be minimized. The front of the furnace shall be constructed of material to prevent distortion.

3.6.3 Hearth. The hearth plates shall be made of materials that will readily conduct heat to the interior of the furnace and shall withstand atmosphere and temperatures encountered in normal service use. The hearth shall be capable of supporting maximum furnace loads under all operating conditions without sagging or warping. The type II, class B and class C furnace shall have hearth plates constructed of silicon carbide. When specified (see 6.2.1), rollers or slide rails shall be provided for moving the charge in and out of the chamber.

3.6.3.1 Shelves. When required, the furnace shall be provided with removable open grid type shelves. Number and arrangement of shelves shall be as specified (see 6.2.1).

3.6.4 Refractory and insulation. Refractory and insulation shall consist of insulating fire brick and block, or preformed insulation selected to ensure long life and minimum heat loss. Overall thickness and quality of the refractory and insulation shall be such that with the chamber temperature stabilized at maximum operating temperature, the outside surface temperature of the type I, class A and class B, and type II, class A furnace shall not exceed 100°F above ambient temperature, or the type II, class B 150°F above ambient temperature, or the type II, class C 200°F above ambient temperature. Ambient temperature is defined at 77 ± 18°F in still air. Powder, shredded or granular insulation shall not be used. When mortar is used, it should be of the same material as the units being cemented together.



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3.6.5 Door. Unless otherwise specified (see 6.2.1), the door shall be manually operated and of the vertical lift type with suitable counter weights. The door shall be fully insulated and faced with a refractory material where exposed to the internal furnace temperature. The door shall be provided with jambs, or equally suitable means for sealing it in the closed position, and shall overlap the brick facing of the furnace not less than 2.5 inches on all four sides.

3.6.6 Air circulating system. Type I, class A and class B furnaces shall be provided with one or more fans of the circulating type, constructed of heat resisting, non-oxidizing alloy suitable for continuous operation at maximum furnace operating temperature. The number of fans and the capacity of the system shall assure adequate circulation of air for uniform temperature control as specified in 3.5.2. Adequate means shall be provided to assure proper cooling for all bearings and components. Fans shall be inter-locked or provided with a safety switch to prevent operation of the furnace in the event of fan failure.

3.6.7 Heating elements. Furnaces having a temperature rating of 2000°F or less shall have heating elements fabricated from a high-resistance alloy containing approximately 80 percent nickel and 20 percent chromium with not more than one percent iron as an impurity, or an equivalent alloy having as good or better characteristics. Furnaces having a temperature rating in excess of 2000°F shall have heating elements of silicon carbide or of an equivalent material having as good or better overall characteristics. Unless otherwise specified (see 6.2.1), a multiple-tap transformer with not less than 8 steps for changing voltage shall be furnished with all heating elements of the silicon carbide type to compensate for the change in resistance. An ammeter or a comparable instrument shall be furnished with furnaces having heating elements of the silicon-carbide or equivalent type to detect changes in resistance. The power density of the heating elements shall not exceed 15 watts per square inch of radiation surface for alloy type elements, 50 watts for silicon-carbide elements operating at 2500°F, and 25 watts for silicon carbide operating at 2750°F.

3.6.7.1 Terminals. The terminals of the metallic heating elements shall be designed to have not less than twice the current-carrying capacity of the heating element. Solderless connections to power leads shall be furnished.

3.6.8 Thermocouples. Furnaces having a temperature rating in excess of 2000°F shall be provided with platinum-rhodium type thermocouples. Furnaces having a temperature rating of 2000°F and lower shall be provided with chromel-alumel type thermocouples. The thermocouples shall be adequately protected from contamination of furnace atmospheres by means of suitable protective tubes. The thermocouples shall conform to ANS C96.1. Unless otherwise specified (see 6.2.1), thermocouple lead wires shall be not less than 25 feet in length. Thermocouple wires shall be of the same material as the thermocouples, or have the same thermoelectric characteristics.



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3.6.9 Instruments. The instruments shall be of solid state design, shall be marked to indicate the particular type thermocouple for which it is calibrated and have automatic reference junction compensation in all instruments using thermocouples as the temperature sensing element. The controllers shall provide thermocouple upscale failsafe protection that will shut off the heat source at approximately 100°F above maximum temperature.

3.6.9.1 Temperature controller. Unless otherwise specified, the furnace shall be provided with an indicating, recording, controlling type instrument having an accuracy of 0.25 percent or better, of full scale reading and capable of maintaining the set control temperature within + 2°F within the operating range of the furnace. When specified, the furnace shall be provided with an indicating non-recording, controlling type instrument with an accuracy as specified (see 6.2.1). The instrument shall be of the time proportioning type control and shall operate a contactor through an intermediate relay system.

3.6.9.2 Control indicator scale. The temperature control indicator scale of controller with recorder shall be approximately the same length as the chart width and the control indicator scale width or non-recording controller shall be not less than 6 inches. The calibrated scale range shall be not less than 100°F above the maximum temperature of the furnace. The increments between successive temperature marks shall be not greater than 20°F, with the spacing of numbered marks not greater than 400°F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

3.6.9.3 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified, the calibrated chart width shall be not less than six inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart width, and not less than seven inches of chart length. When required, the chart width shall be not less than 9 inches and provisions provided for changing chart speed to 4 and 6 inches per hour as specified (see 6.2.1).

3.6.9.4 Excess temperature controller. When specified (see 6.2.1), the furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry. The controller shall provide for selecting the over-temperature control point and shall function to shut-off the heat source to the furnace when the over-temperature control point is reached. The excess temperature controller shall provide for furnace shut-down in the event of thermocouple failure, or failure of other components in the circuit.



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In case the furnace is shut-down by the excess temperature control system, it shall sound an audible alarm and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

3.6.9.5 Program control. When required, a temperature programmer shall be furnished and integrated with the furnace control system to provide temperature control in accordance with a preselected program as specified (see 6.2.1).

3.6.9.6 Instrument panel. Unless otherwise specified (see 6.2.1), one floor mounted type instrument panel conforming to JIC EGP-1 shall be provided with each furnace for mounting all instruments and accessory controls. The control accessories shall be mounted in a front opening NEMA type 12 enclosure with a disconnect switch. A separate enclosure shall be provided for mounting the power contactor individually and separate from the instrument panel. The panel shall provide for semi-flush or flush mounting of all instruments. When specified (see 6.2.1), the instrument panel shall be provided with shock-absorbing mounts. Signal lights shall be provided to indicate the furnace cycle performance.

3.6.9.7 Vibration and tilt requirements. When specified (see 6.2.1), indicating, recording, and control instruments shall retain their accuracy within the tolerances specified when:

a. Exposed to a vibration of 0.030 inch excursion at 18 cycles per second.
b. Operated in tilted positions 15 degrees from the normal vertical position in any direction.

3.7 Time totalizing meter. When specified (see 6.2.1), the furnace shall be provided with a meter to measure actual operating time of the heating elements. The time totalizing meter shall be of the nonresetting type and shall have a range of 0 to 100,000 hours in increments of 1 hour and 99999 shall be the maximum readout. The least significant digit on the meter readout shall be 1 hour. Lesser increments are not acceptable. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time. The meter shall be designed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the equipment.

3.8 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.



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3.9 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the nameplate shall contain the model number of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
Power input characteristics and ratings
Inside dimensions (width, length, height - in inches)
Maximum temperature degrees F
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code
U S

3.10 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.11 Workmanship. Workmanship of the furnace and accessories shall be commensurate with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).



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4.3 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.5 and all tests in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1 and 4.6.2 and the inspection in 4.7. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical and safety compliance, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. All tests shall be conducted at an ambient temperature of $77^{\circ}F \pm 18^{\circ}F$. Test instruments and sensing elements used to perform the test specified for the furnace and instruments should have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous three months to read accurate within $\pm 2^{\circ}F$.

4.6.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANSI C96.1. The instruments shall be tested through their operating range.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of JIC Standards.

4.6.3 Electromagnetic interference control testing. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.4 in accordance with table II of MIL-STD-461.

4.6.4 Tilt test. When specified (see 6.2.1), the tests specified in 4.6.1 shall be repeated with the indicating and recording instruments at an angle of 15 degrees in any direction from the normal vertical position. The instruments shall operate and function within the tolerances specified.



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4.6.5 Vibration tests. When specified (see 6.2.1), the tests specified in 4.6.1 shall be repeated with the indicating and recording instruments being subjected to a vibration of 0.030 inch excursion at 18 cycles per second for a period of 15 minutes each in the lateral and transversal horizontal planes and in the vertical plane. The instruments shall function within the prescribed tolerances other than for minor adjustments.

4.6.6 Performance test. The furnace shall be prepared for operation and operated in accordance with the manufacturer's handbook of instructions until the refractory and insulation has cured. With the temperature stabilized at the temperature specified in 3.5.3, the furnace shall be loaded with a batch type charge of iron or steel equivalent to not less than the recovery rate specified for the type and class furnace under test. The recovery cycle shall be recorded and at the end of one hour, the furnace temperature shall be within 10 degrees F. of the initial temperature. Each safety device, such as loss of thermocouple circuit and fan failure safety device shall be tested by simulating a failure that normally would cause the system to operate. Each device or system shall perform its intended function.

4.6.7 Temperature uniformity test. The furnace shall be charged as specified in 4.6.6 and a temperature survey performed. A minimum of nine test locations shall be used, with one thermocouple located in each corner of the heating chamber of type I furnaces. For type II furnaces, one thermocouple shall be located in each corner of the effective work zone. One thermocouple shall be located within three inches of each control thermocouple. For furnace with a chamber volume of less than 10 cubic feet, the test may be performed with a minimum of three thermocouples, with one located at the front, center, and rear of the test zone. The survey shall be performed at the minimum and maximum temperature of type I furnaces and the type II, class A furnace, and at the minimum and at 2150°F for the type II, class B and class C. From the time the heating cycle is initiated until the thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the temperature range being surveyed. After thermal equilibrium is reached, readings shall be taken at 5 minutes intervals for 30 minutes. The maximum temperature variation between all points under test shall not exceed $\pm$ 10°F, and the maximum temperature variation between the furnace instrument control set point and all points under test, shall not exceed $\pm$ 25°F.

4.6.8 Chamber surface temperature test. The chamber temperature shall be stabilized at maximum operating temperature, and the surface temperature of the chamber and door checked at random locations. The surface temperature shall not exceed the limits specified in 3.6.4, except within 3 inches of through-fastening devices. The type II furnaces shall be unloaded before increasing the temperature above the melting point of the charge.



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4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing, and marking shall be in accordance with MIL-F-3296. Level of preservation and packing shall be as specified (see 6.2.1).

5.2 Marking

5.2.1 Civil agencies. In addition to the marking required by the contract or order (when specified, see 6.2.1) marking shall be in accordance with Federal Standard No. 123.

6. NOTES

6.1 Intended use. Furnaces covered by this specification are for use primarily in heat-treating shops for hardening, tempering, normalizing, and annealing metal parts. Furnaces covered by this specification are not intended for atmosphere control use.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2.1).
c. Size required; specify width, length, and height (see 1.2.2).
d. First article, when required (see 3.1).
e. Bench mounted, if required (see 3.2).
f. Specify additional safety and health requirements (see 3.2.3 and 6.5).
g. Painting, if different (see 3.3.5).
h. Power supply, specify voltage, phase, cycle (see 3.4.1).
i. Suppression of radio interference, when required (see 3.4.4).
j. Hearth rollers or slide rails, when required (see 3.6.3).
   NOTE: Chamber working height will be decreased by a dimension equal to the thickness of the rails or height of the rollers in some designs.
k. Number and arrangement of shelves, when required (see 3.6.3.1).
l. Door, if different (see 3.6.5).



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m. Transformer, if different (see 3.6.7).
n. Thermocouple lead wire length, if different (see 3.6.8).
o. Non-recording temperature controller and control accuracy, when required (see 3.6.9.1).
p. Recorder chart width and provisions for changing chart speed. Specify chart width and chart speed required (see 3.6.9.3).
q. Excess temperature controller, when required (see 3.6.9.4).
r. Program control and program requirements when required (see 3.6.9.5).
s. Instrument panel, if different, specify if floor mounted panel is not required, or if other than JIC type panel is acceptable (see 3.6.9.6).
t. Instrument panel shock-absorbing mounts, when required (see 3.6.9.6).
u. Instrument panel, when required to operate under vibration and tilt conditions (see 3.6.9.7).
v. Hour meter, when required (see 3.7).
w. Repair parts, when required (see 3.8).
x. Nameplate, if different (see 3.9).
y. First article inspection, if different (see 4.3).
z. Quality conformance inspection, if different (see 4.4).
aa. Tilt test, when required (see 4.6.4).
bb. Vibration test, when required (see 4.6.5).
cc. Level of preservation and packing required (see 5.1).
dd. Civil agency special marking, when required (see 5.2.1).

6.3 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, tests and approval of the first article.

6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.6 Mercury prohibited. When the use of mercury or its compounds in the heater and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.



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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL, EL, SM
Navy - SH
Air Force - 99
DLA - GS

User activities:
Army - AR, AV, EL
Navy - AS, YD

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

Preparing activity:
DLA - IP

Project Number:
3424-0089

U.S. GOVERNMENT PRINTING OFFICE : 1979 - 281-172/1198

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 1.00cents each.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

W-F-785E, Furnaces, Heat-Treating, Electric, Horizontal Box Type

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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